Intelligent Systems

Card Vision

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Application Idea

This application acts as an **intelligent assistant** for **card games** by automatically recognizing and tracking cards in a player's hand and on the game board, providing real-time support, strategic recommendations, and statistical insights.

- Real-time card recognition using computer vision.
- Move suggestions based on game rules and strategy analysis.

Use Case Scenarios

- **Poker Assistant**
  - Real-time card recognition from the game table
  - Hand evaluation and suggestions through the analysis module
- **Game Digitization**
  - Automatic capture and storage of game states
  - Digital replay, archiving, and post-game review
- **Statistical Analysis**, **Decision Support**
  - Collection of card data across multiple games
  - Analysis of game patterns, player strategies, and statistics
  - Data acquisition for ML-based decision-support systems

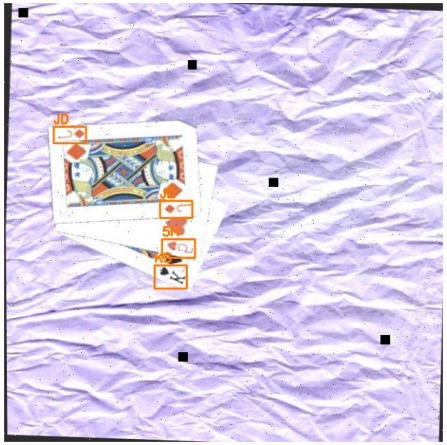
Proposed Intelligent System

The proposed IS is composed of multiple components:

- Image acquisition module
- Card detection module — based on a YOLO OD model
- Same-Card Recognition module
- Poker analysis module — processes the recognized cards and derives information about the current poker hand
- User interface/output module

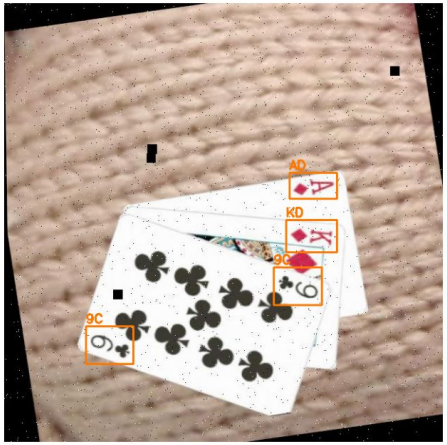
Datasets

The found card games dataset are partially or fully **synthetic**.



2 different datasets has been merged:

- Kaggle Dataset
- Robflow Dataset



Data have been augmented as well, exploiting YOLO augmentation pipeline.

Object Detection Model

Development of the central **object detection model** for the proposed Intelligent System

- Comparison of different ***YOLOv11*** variants:
  - *Nano, Small, Medium, Large*
- All models are initialized from pretrained weights, then fine-tuned on the target dataset
- Models are evaluated through **cross-validation** to ensure a statistically robust comparison

Cross Validation Strategy

Cross-Validation enables a statistically robust comparison:

A **5-fold Cross Validation** (*K=5*) strategy is employed:

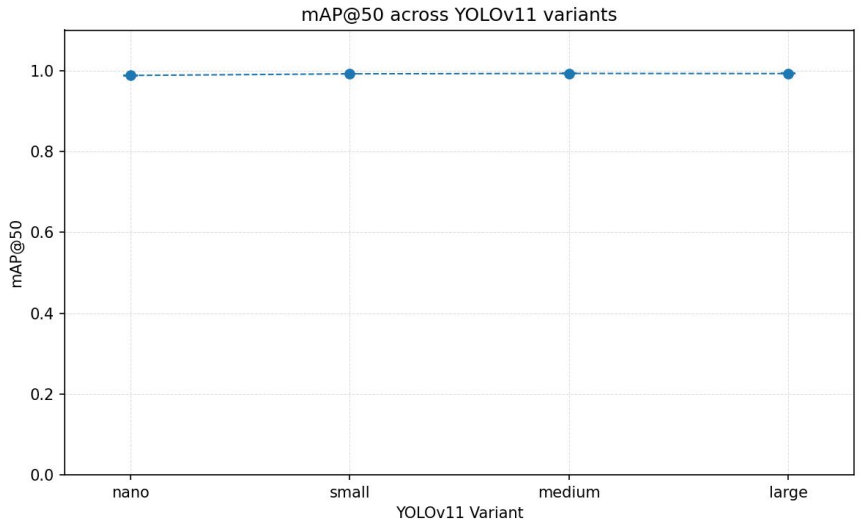
- At each iteration one fold is used as *test set*
- The remaining K-1 fold are used for training and further splitted into *train set* and *validation set* (80/20)

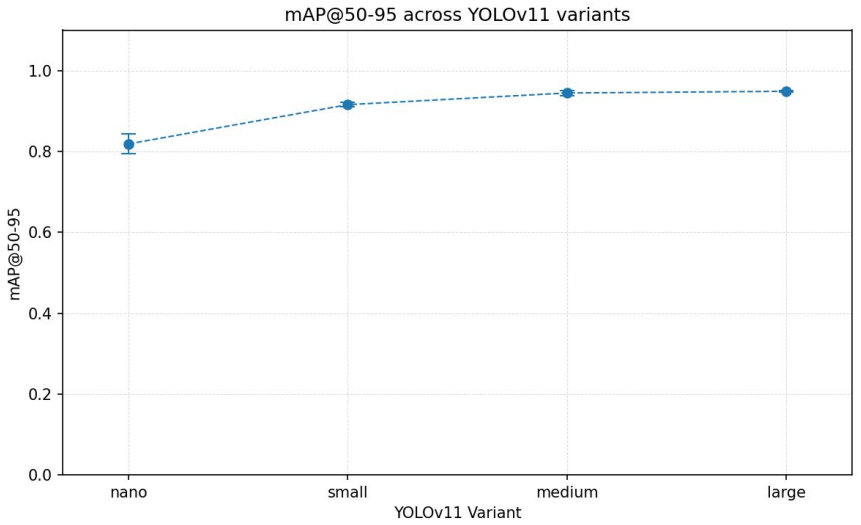
Performance metrics are finally averaged across the five folds to compare and select the best variant

Model Comparison (I)

Playing card detection task is **straightforward**:

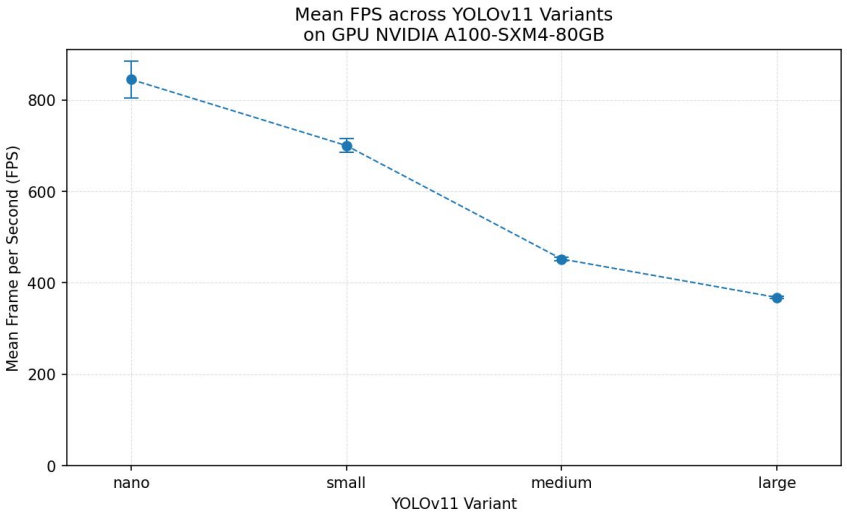
- playing cards are well-defined rectangular white object
- target bounding boxes are aligned with their corners

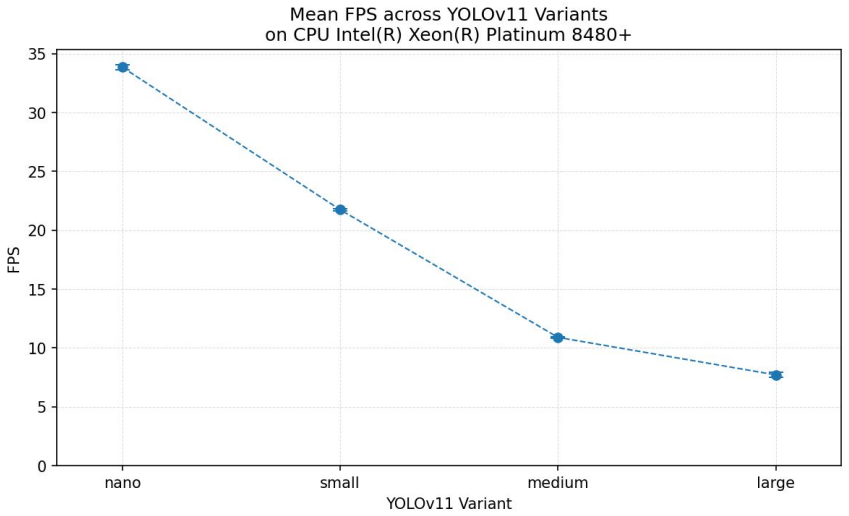




Model Comparison (II)

The maximum achievable frame rate (FPS) decreases as the model complexity increases

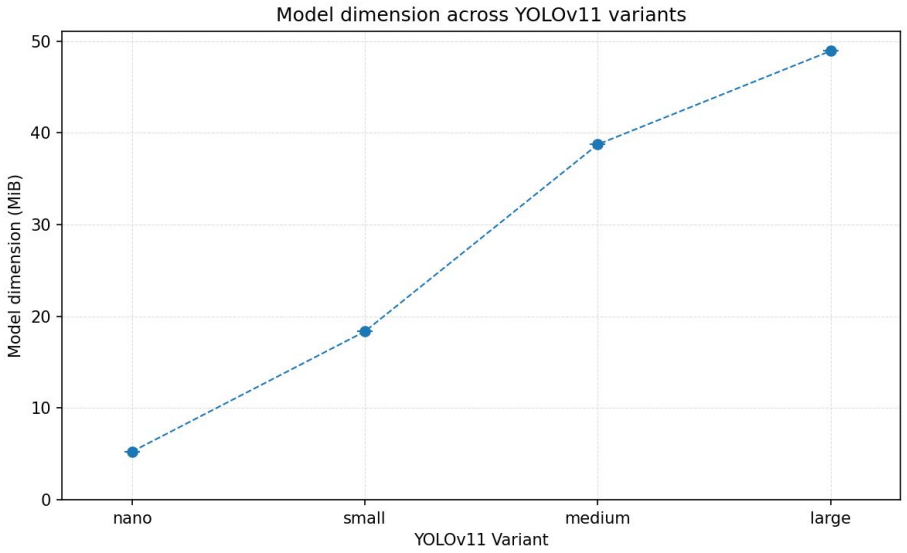




Model Comparison (III)

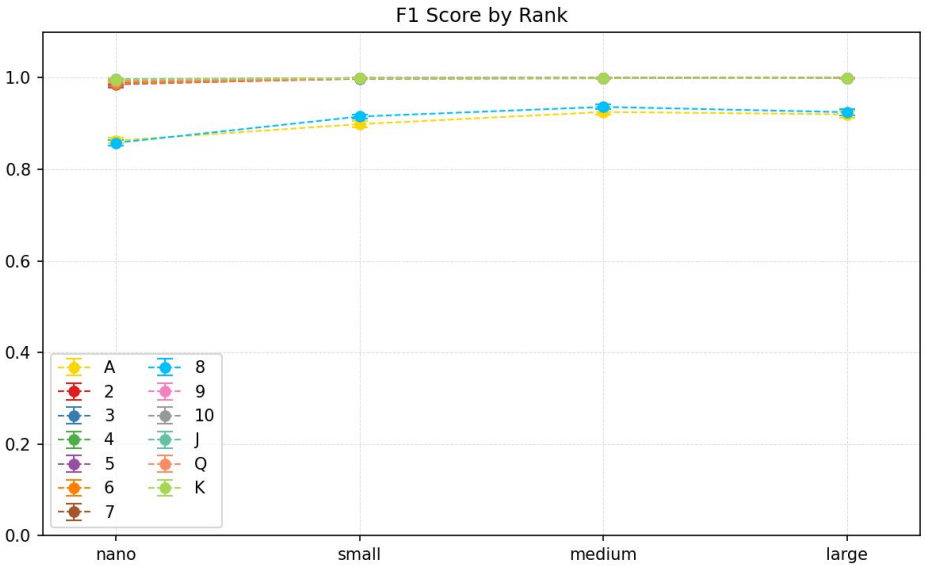
Increasing the model size results in larger model dimensions, roughly:

- *Nano*: 5MiB
- *Small*: 20MiB
- *Medium*: 40MiB
- *Large*: 50MiB



Model Comparison (IV)

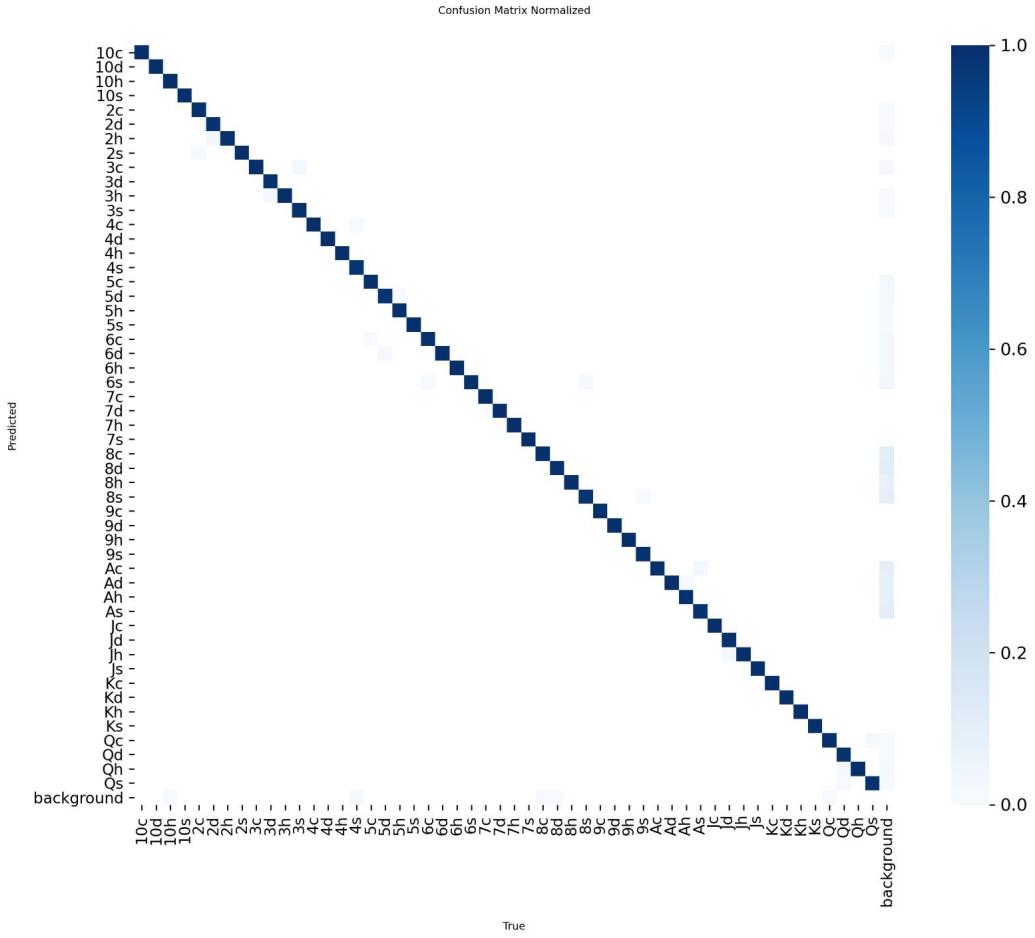
Ace and **Eight** ranks consistently achieve lower scores, while the remaining ranks exhibit very similar, higher performance



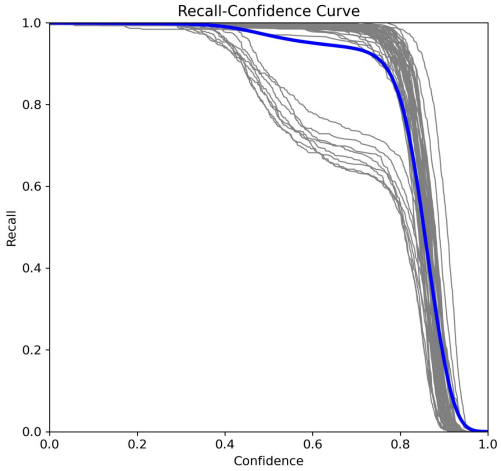
Model Selection (I)

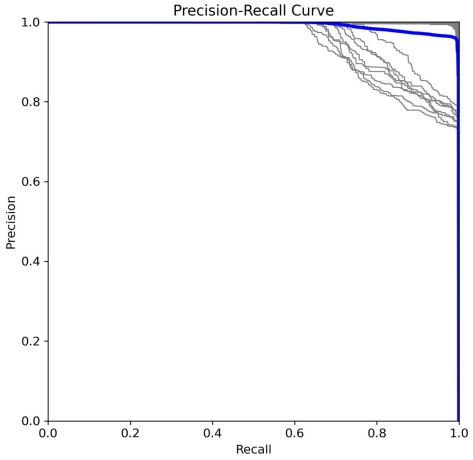
We selected the **small** variant, deemed the best trade-off between accuracy and computational efficiency

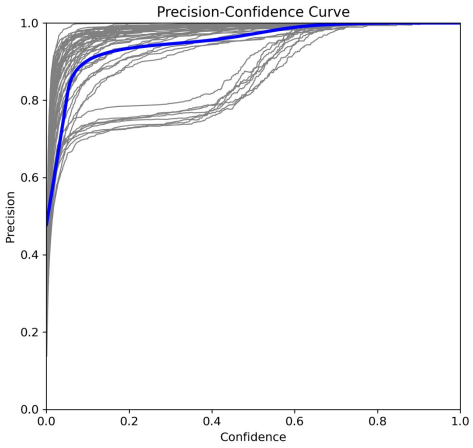
- Significant improvement in *mAP@50-95* compared to *Nano*.
- Minor increase in model siz and inference time

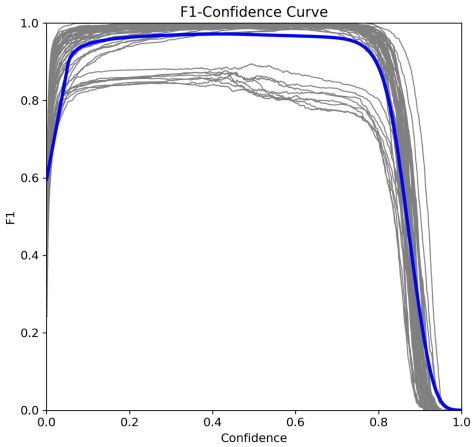


Model Selection (II)





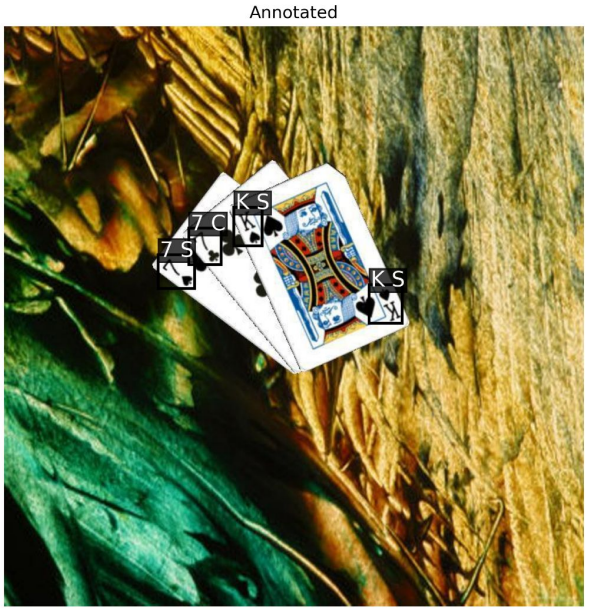




Same Card Recognition Problem (I)

Introduction

- The object detection model detects the card identifying elements (**rank and suit**), rather than the whole card
- Hence a single visible card generates two distinct detections with the same class label

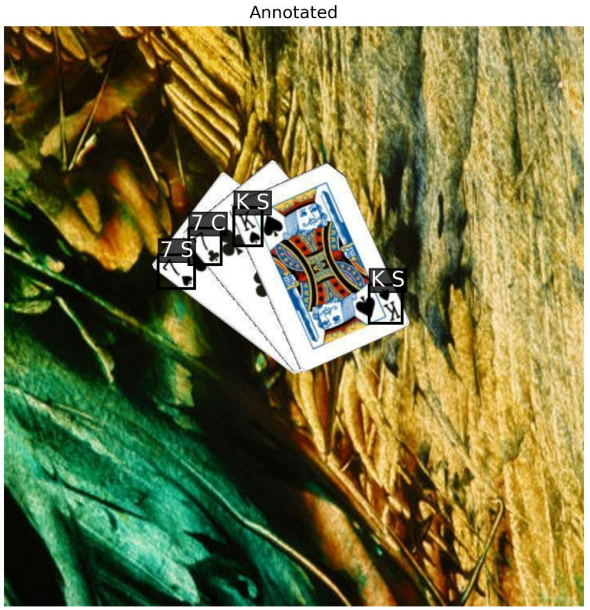


Same Card Recognition Problem (II)

Motivation

- **Multi-deck** games present multiple instances of the same card
- The IS must distinguish and keep track of individual card instances in the scene

Note: Single-deck games don't require card instance disambiguation.



Same Card Recognition Problem (III)

Proposed approach

For an image, consider all detected bounding boxes belonging to the same class

- If 0 or 1 BBox is detected for a class, no ambiguity occurs
- If 2+ BBoxes are detected, all possible pairs are evaluated

A **metric**, defined over two BBoxes, is developed to determine whether two same-class bounding boxes belong to the same physical card

Same Card Recognition Problem (IV)

Proposed metric

- *Idea*: For a completely visible card, its BBoxes dimension and the relative position are related to the dimension of the card, and scale with the card relative size
- *Metric*: ***DoA*** – distance between two bounding boxes centers over the square root of the average of their areas

$$DoA(bb_1, bb_2) = \frac{\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}}{\sqrt{\frac{A_1 + A_2}{2}}}$$

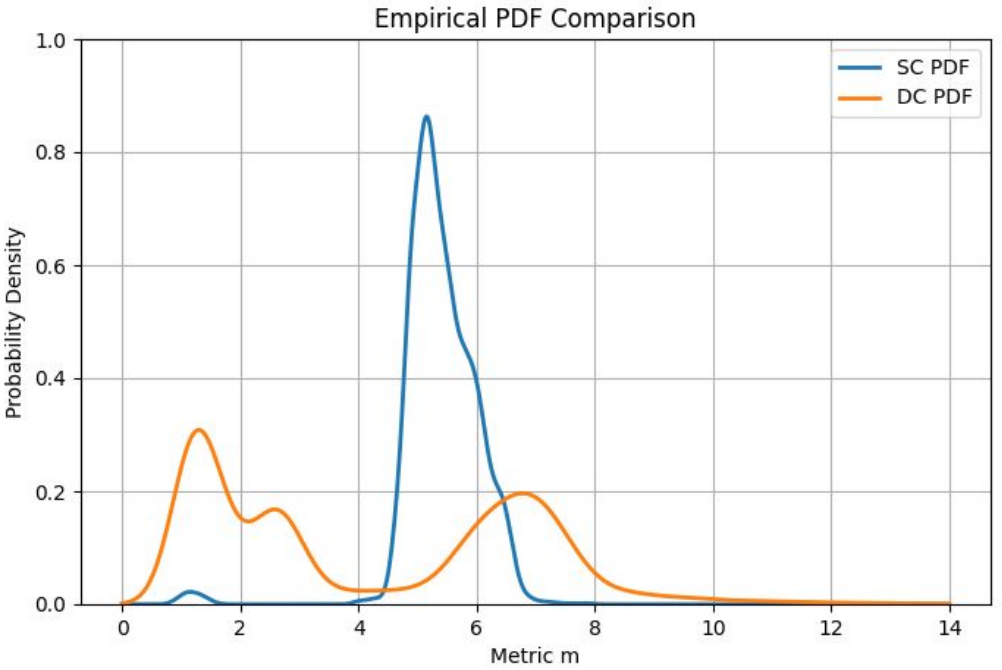
Same Card Recognition Problem (V)

Metric evaluation

An empirical Probability Density Function (PDF) of the *DoA* metric is computed for:

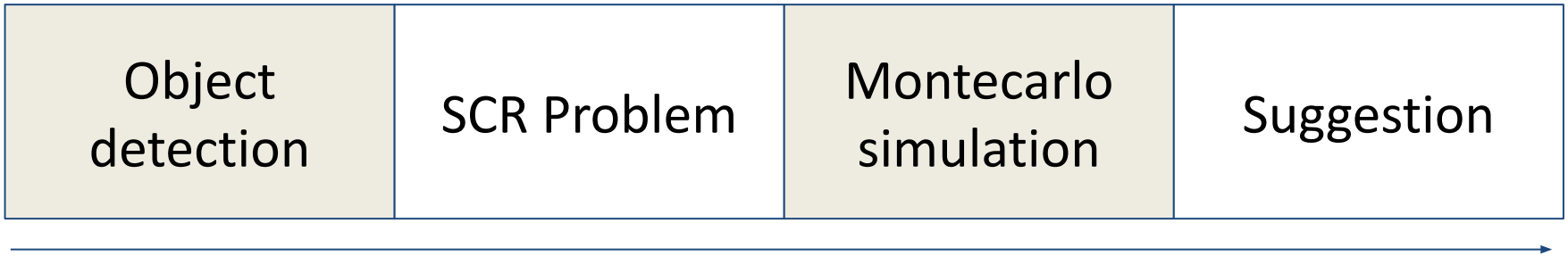
- Same-Card pairs
- Different-Card pairs (for comparison)

Same-Card pairs show DoA values within the **4.5 - 6.7** range



Intelligent System I

Intelligent System's pipeline



Texas Hold'em Poker Assistant

- Reconstructs the game state from input images.
- Applies Monte Carlo simulation to estimate win probabilities.
- Provides decision support to the player.

IS II - Game State Reconstruction

- Input:
  - Player cards (mandatory)
  - Community cards (optional)
- Automatic identification of the current game stage:
  - Pre-Flop, Flop, Turn, River
- Conversion of detected cards into a structured game state using YOLO model
- Validation of the configuration:
  - No duplicated cards
  - Valid Texas Hold’em state

IS III - Montecarlo Simulation

Random simulation of possible future game scenarios:

- Completion of missing community cards
- Generation of opponents' hands

Estimation of probabilities by repeating this a lot of times:

- $P_{win} = N_{win} / \#$ iterations

Action recommendation (default values) based on P_win:

-

RAISE	**CALL**	**FOLD**
$P_{win} > 0.75$	$0.75 >= P_{win} > 0.45$	$P_{win} <= 0.45$

Conclusion and Future Works

Conclusion

- An O.D. model for playing cards has been developed
- The SCR problem was addressed and a solution was proposed
- A prototype has been developed for the proposed IS

Future Works

- Evaluate the model on a real-world (non-synthetic) dataset
- Extend the prototype into a complete real-time game analysis system, introducing object tracking and live gameplay analysis